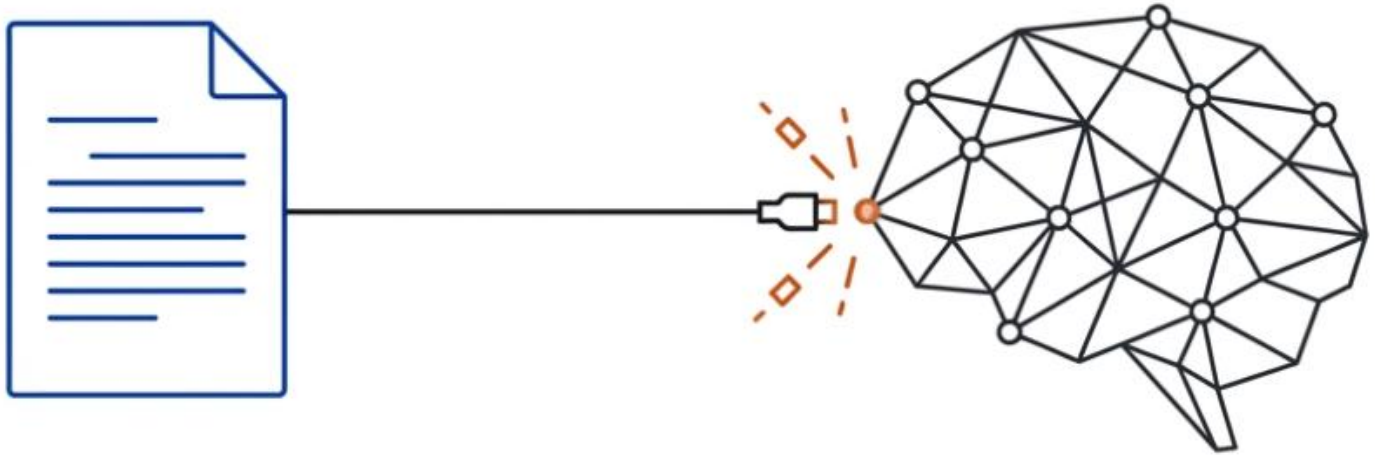
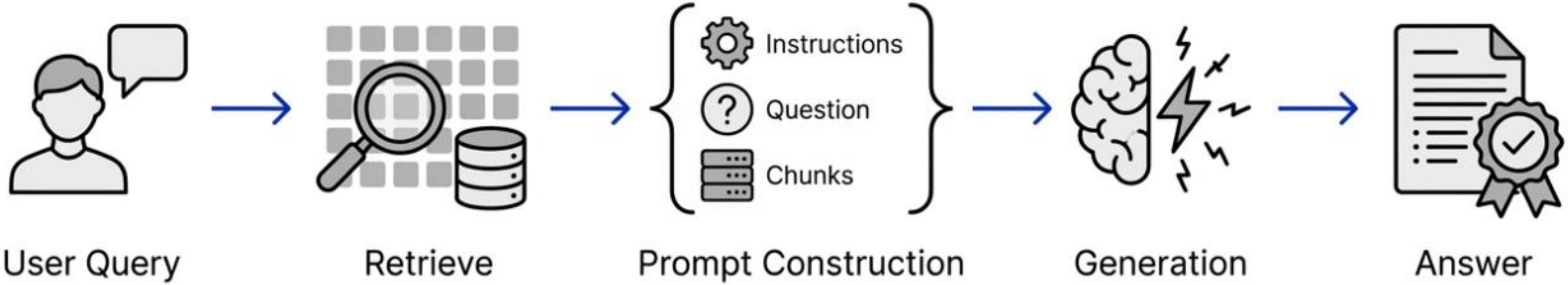


# Search Foundations

Vector Embeddings -  Sparse and Dense and Hybrid, Scoring.
Metrics: Precision@K, Recall@K, MRR

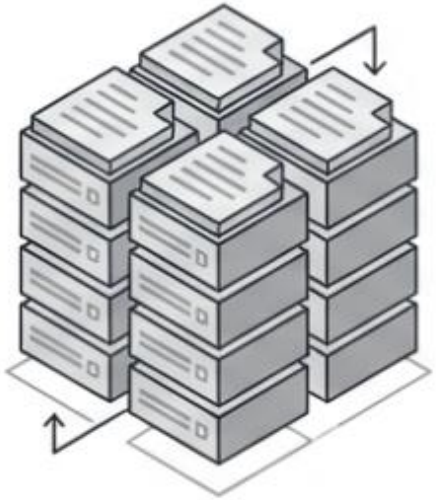
https://tinyurl.com/rag-augmented

# The End-to-End RAG Flow



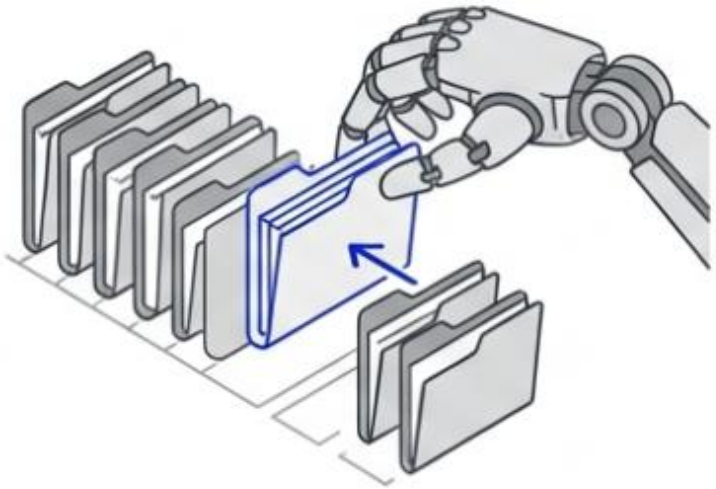

This flow turns a creative writing machine into a grounded research assistant.

## The Store
(The Library)



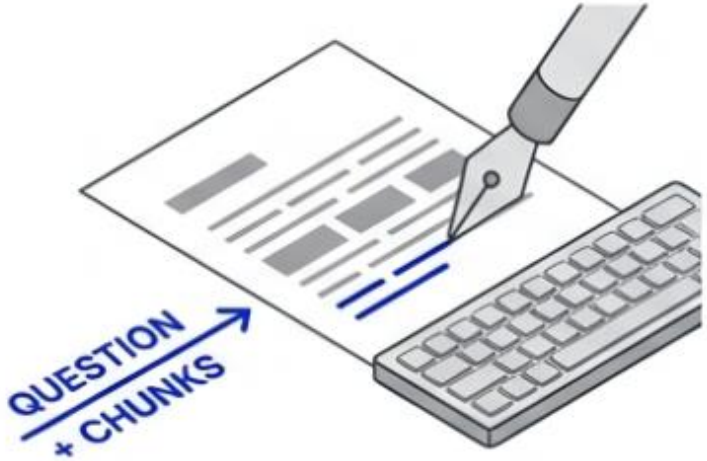
Holds your documents in a retrieval-friendly form.

## The Retriever
(The Librarian)



Fetches the top relevant chunks based on a query.

## The Generator
(The Writer)

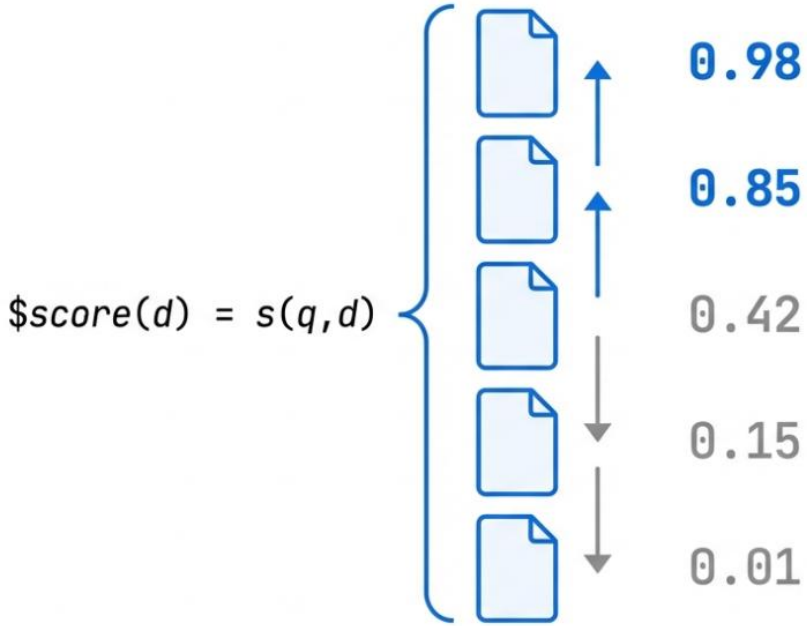


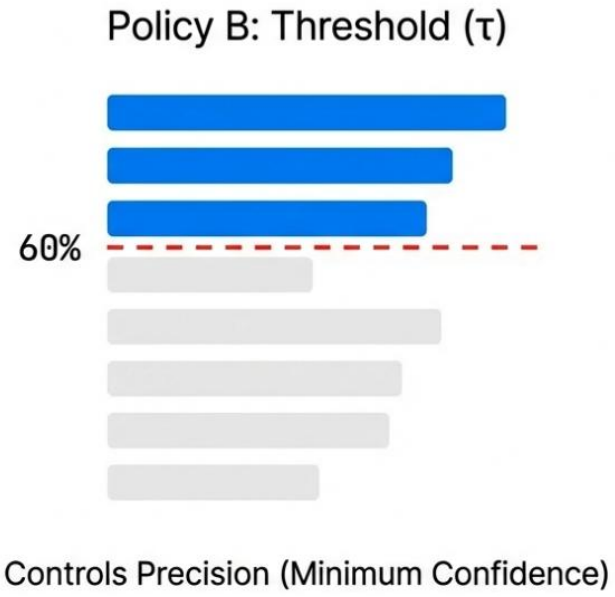

Reads the question + chunks to synthesize an answer.

# Retrieval

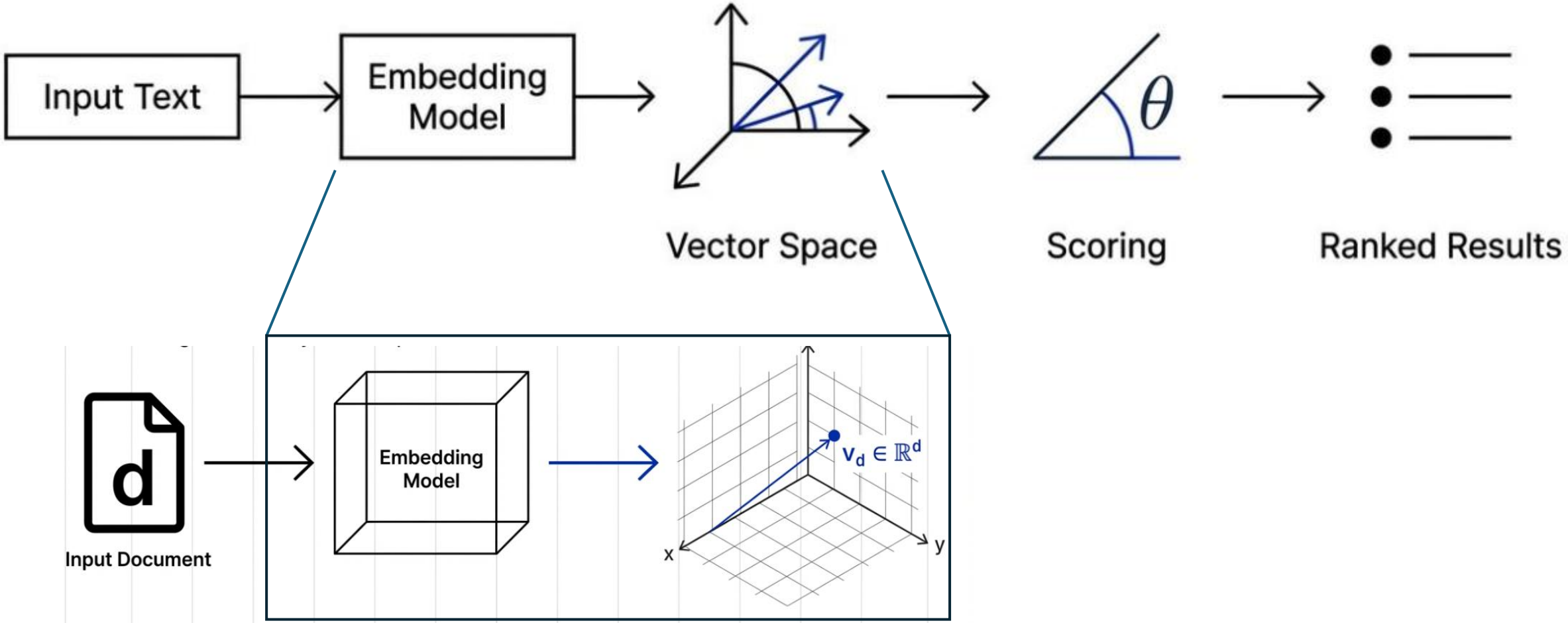
User Query ($q$)
refund policy for damaged item
Transformation
Input: Massive Store ($S$)
Output: Grounded Context ($D_k$)

# Scoring Relevant Documents





# Representing Text (Document and Query)




Retrieval shifts from matching keywords to finding the most similar vector in space.

https://projector.tensorflow.org/

# Scoring: Dot Product

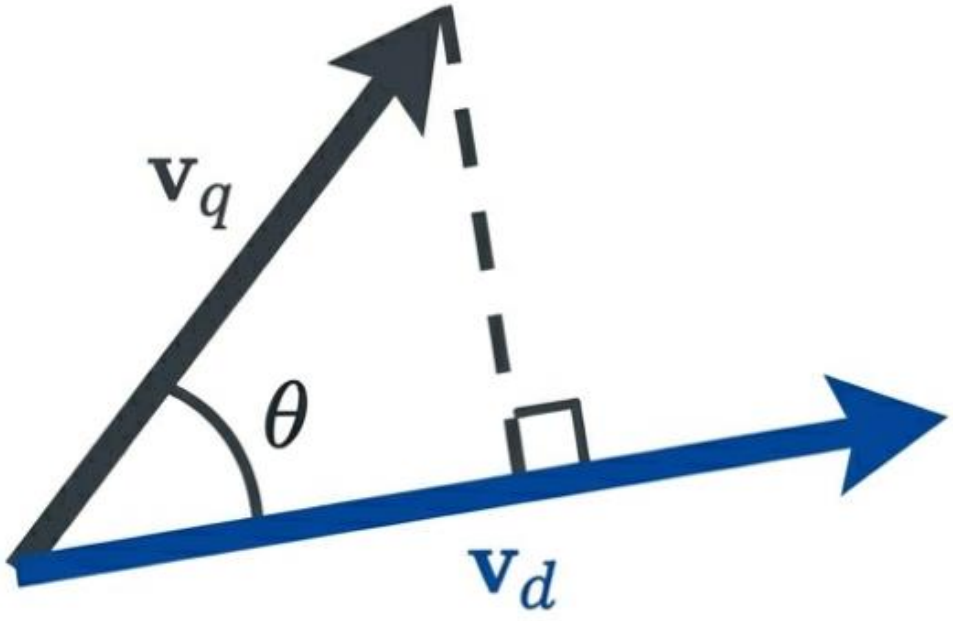
$$\mathbf{v}_q \cdot \mathbf{v}_d = |\mathbf{v}_q||\mathbf{v}_d| \cos(\theta)$$

**Mechanics:**

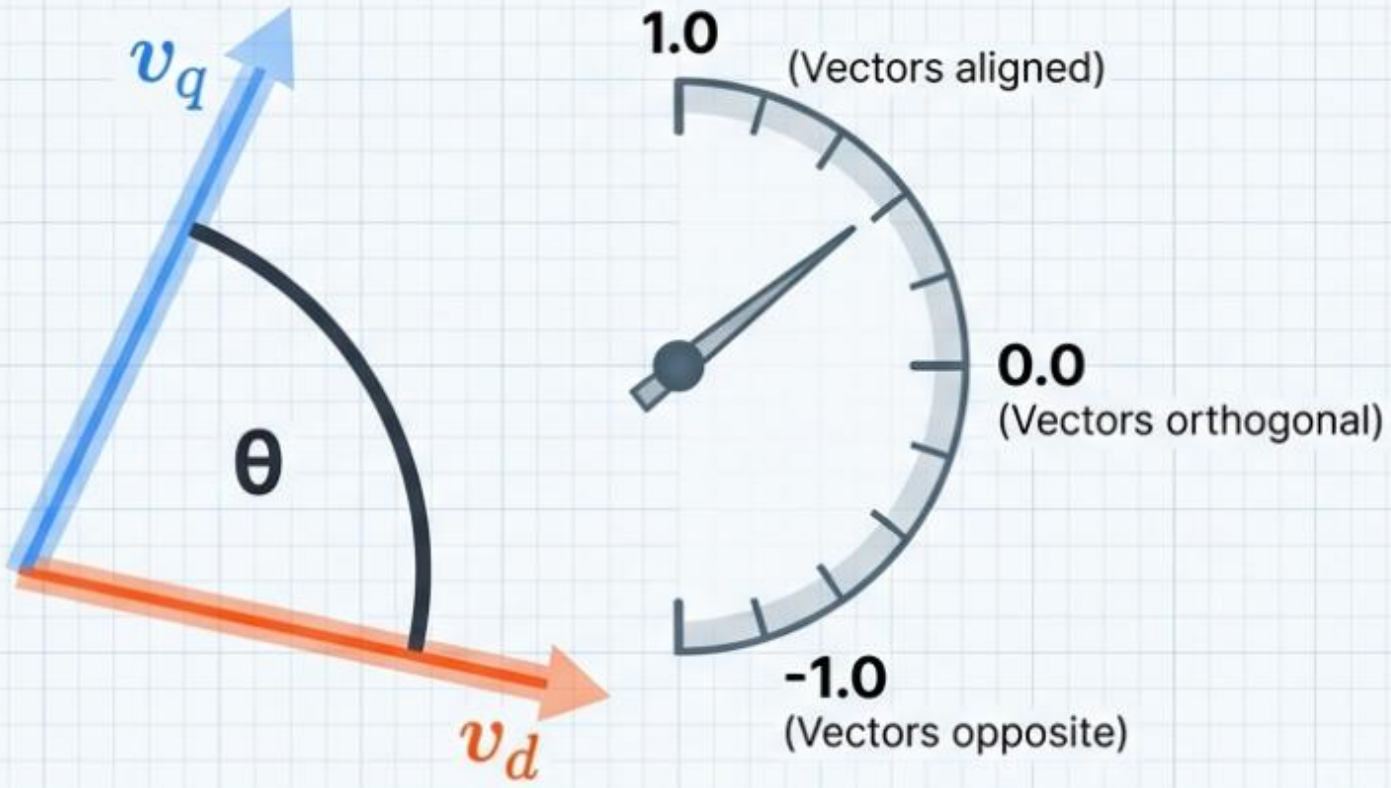
- Score increases with Alignment (small θ)
- Score increases with Magnitude (large |v|)

**Implication:**

Relevant docs align with the query, but high confidence/importance (magnitude) boosts the score.



# Scoring: Cosine Similarity




$$\cos(\theta) = \frac{\boldsymbol{v}_q \cdot \boldsymbol{v}_d}{||\boldsymbol{v}_q||\boldsymbol{v}_d|}$$

**THE NORMALIZER:**
Divides by the product of magnitudes to isolate direction.

# Lexical Search

**Document 1 (Positive Match)**

Laptop battery is not charging. Check the power adapter and the charging port.

**Document 2 (Weak Match)**

If the laptop will not turn on, connect the charger and try a hard reset.

**Document 3 (Negative Match)**

Refund policy: request a refund within 30 days for a damaged item.

---

**User Query:** `laptop battery not charging`

# Step 1: Create a set of terms (vocab)

**Input:**

Laptop battery is not charging...

**Apply Filter**

**Output Tokens:**

Term Frequency > 1

[laptop, battery, not, **charging**, check, power, adapter, **charging**, port]

The complete universe of known terms (N=22).

$$V = \left\{ \begin{array}{cccc} \text{adapter} & \textbf{battery} & \text{charger} & \textbf{charging} \\ \text{check} & \text{connect} & \text{damaged} & \text{days} \\ \text{hard} & \text{item} & \textbf{laptop} & \text{not} \\ \text{on} & \text{policy} & \text{port} & \text{power} \\ \text{request} & \text{refund} & \text{reset} & \text{try} \\ \text{turn} & \text{within} & & \end{array} \right\}$$

# TF Matrix (Term Frequency)

$$tf(t,d) = f(t,d)$$

| Term | D1 | D2 | D3 |
|---|---|---|---|
| adapter | 1 | 0 | 0 |
| battery | 1 | 0 | 0 |
| charging | 2 | 0 | 0 |
| laptop | 1 | 1 | 0 |
| not | 1 | 1 | 0 |
| refund | 0 | 0 | 2 |

**tf(t,d)** = count of term **t** in doc **d**

# IDF: Common Words are poor signals

$$idf(t) = \log\left(\frac{N}{df(t)}\right)$$

Rare words are rewarded. Common words are penalized.

## Common Terms

Terms appearing in 2/3 documents
(e.g., ‘laptop’, ‘not’)

```
idf = log(3 / 2) ≈ 0.4055
```

Low Weight

## Rare Terms

Terms appearing in 1/3 documents
(e.g., ‘battery’, ‘charging’)

```
idf = log(3 / 1) ≈ 1.0986
```

High Weight

# TF-IDF

$$w(t,d) = tf(t,d) \cdot idf(t)$$

Term: “charging” TF (2) × IDF (1.0986) = **2.1972**

Term: “battery” TF (1) × IDF (1.0986) = 1.0986

Term: “laptop” TF (1) × IDF (0.4055) = 0.4055

$$\begin{array}{c} \text{D1} \\ \begin{bmatrix} 2.1972 \\ 1.0986 \\ 0.4055 \end{bmatrix} \end{array}$$

# Obtaining Similarity

| | | | |
|---|---|---|---|
| 1 | **Document 1** | | Score `0.7129` - **PASSED** |
| 2 | **Document 2** | | Score `0.0670` - FAILED (Below Threshold) |
| 3 | **Document 3** | | Score `0.0000` - FAILED |

# BM25

Term Importance Weight

$$\mathrm{BM25}(q,d) = \sum_{q} \mathrm{idf}(t) \cdot \left[ \frac{\mathrm{tf}(t,d) \cdot (k1+1))}{\mathrm{tf}(t,d) + k1 \cdot \left(1 - b + b \cdot \frac{||d|}{\mathrm{avg}dl}\right)} \right]$$

Raw Frequency Count

Document Length Penalty

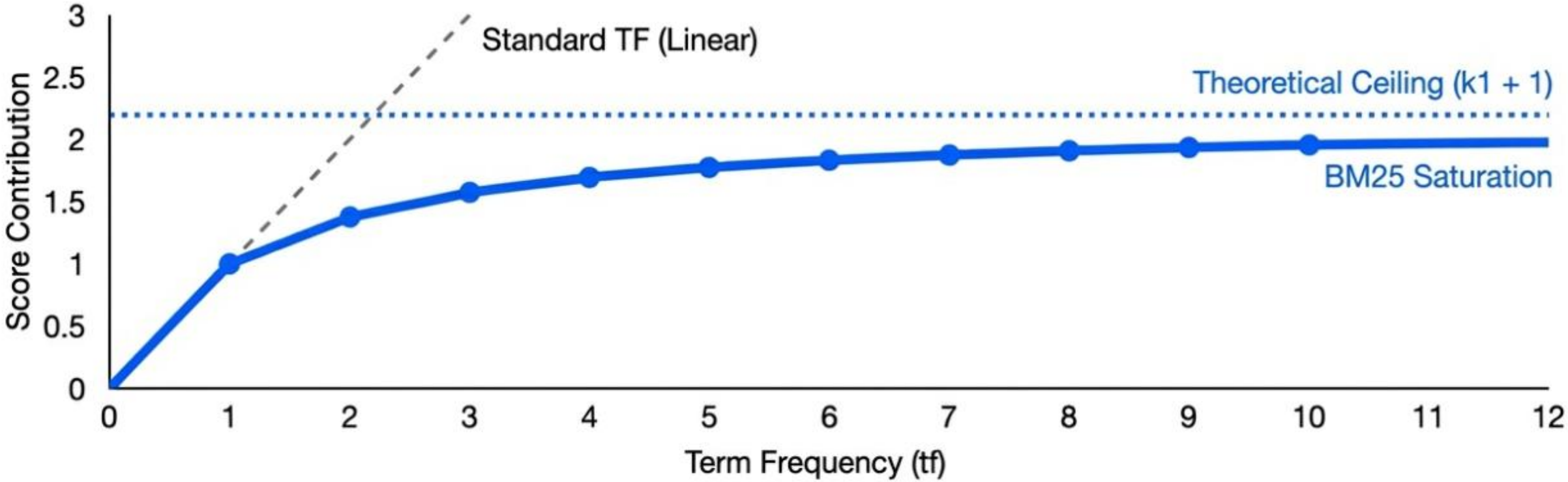
**Parameters**

Saturation (k1) = 1.2

Length Norm (b) = 0.75

# TF Saturation – k1

Why repeating a word 100 times doesn't make it 100x relevant



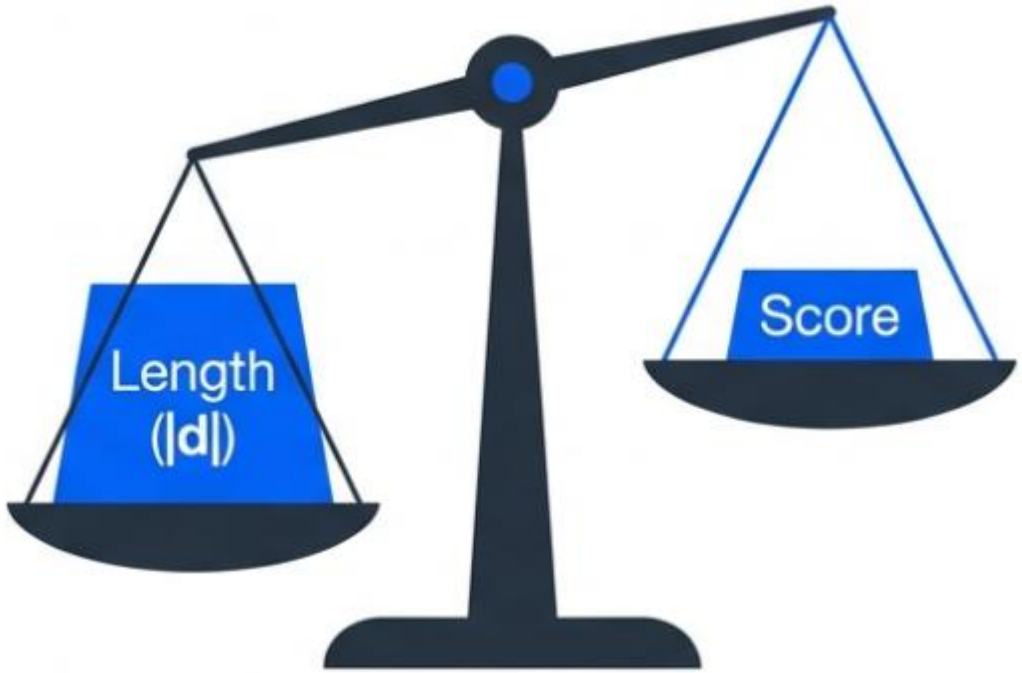

The k1 factor (1.2) enforces diminishing returns.

# Length normalization

The Factor **K**:

$$K = k1 \cdot \left(1 - b + b \cdot \frac{|d|}{\text{avg}dl}\right)$$

Interpretation: As document length **|d|** increases, **K** increases, reducing the final score.



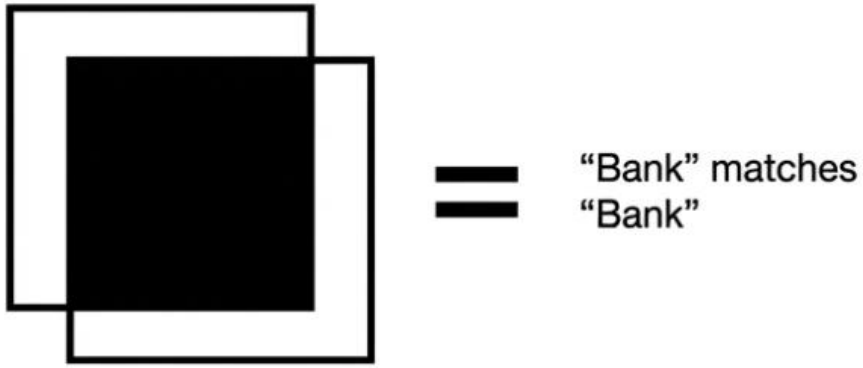

Average Length (**avgdl**) = 8.67

If **|d|** > 8.67, the score is penalized.
If **|d|** < 8.67, the score is boosted.

# Semantic Search

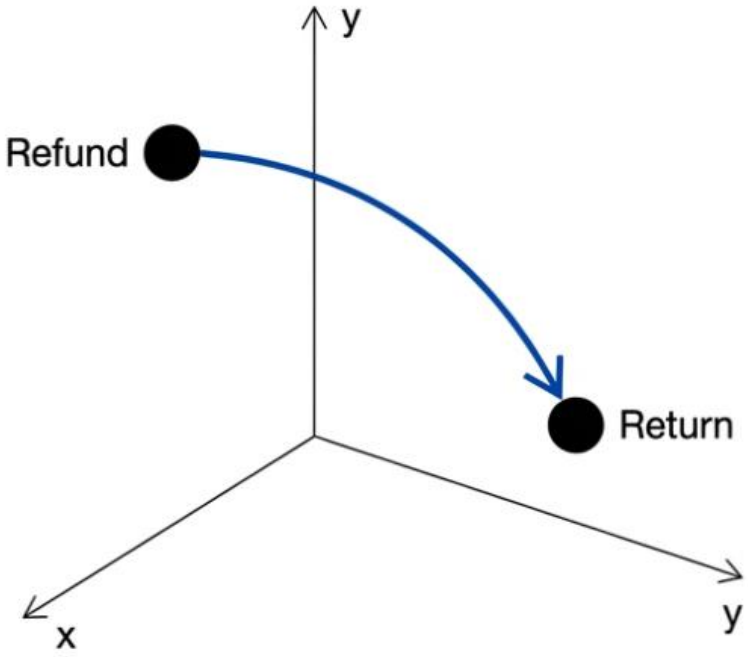
## Matching Words vs. Matching Meaning

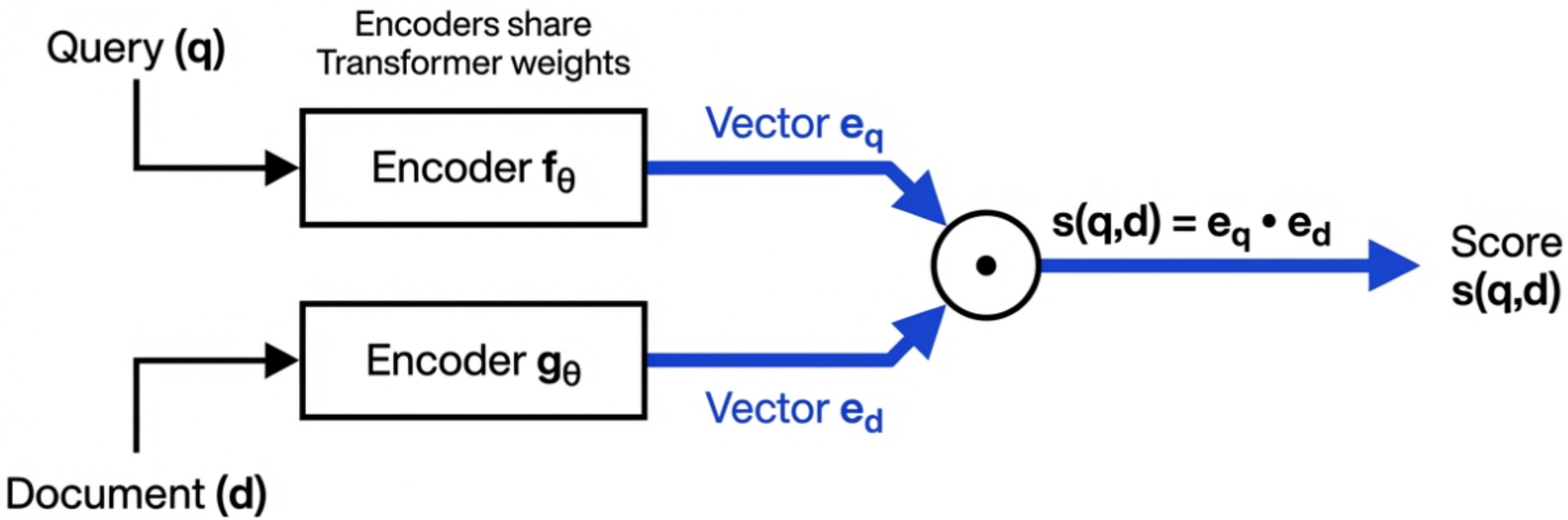
### LEXICAL / TF-IDF



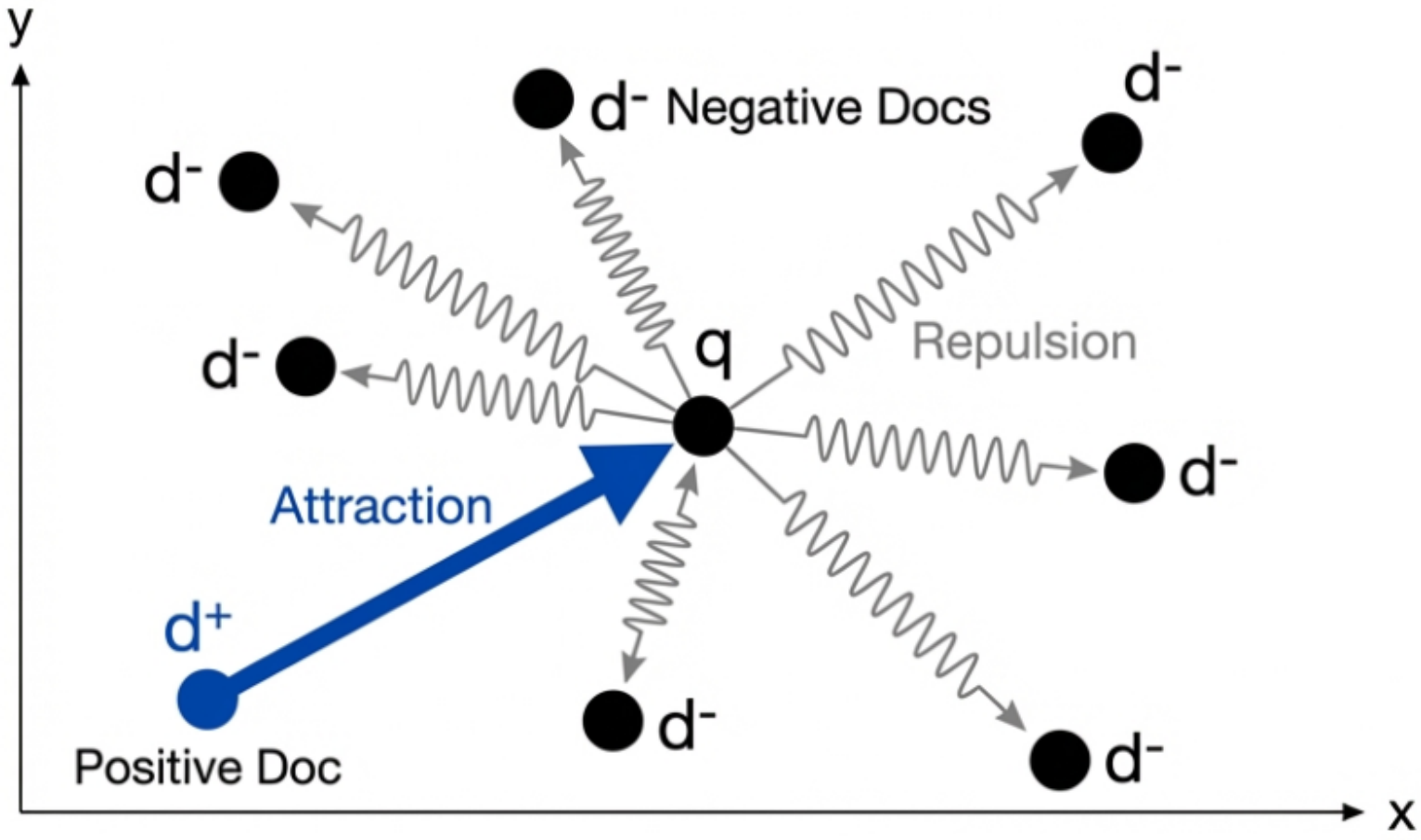

Lexical methods match tokens. Semantic similarity matches meaning by mapping text to a dense vector embedding, capturing relationships even when vocabulary differs.

### SEMANTIC / DENSE




Query (q)
Encoders share
Transformer weights
Encoder $f_\theta$
Vector $e_q$
$s(q,d) = e_q \bullet e_d$
Score
s(q,d)
Encoder $g_\theta$
Vector $e_d$
Document (d)

# Learning Objective

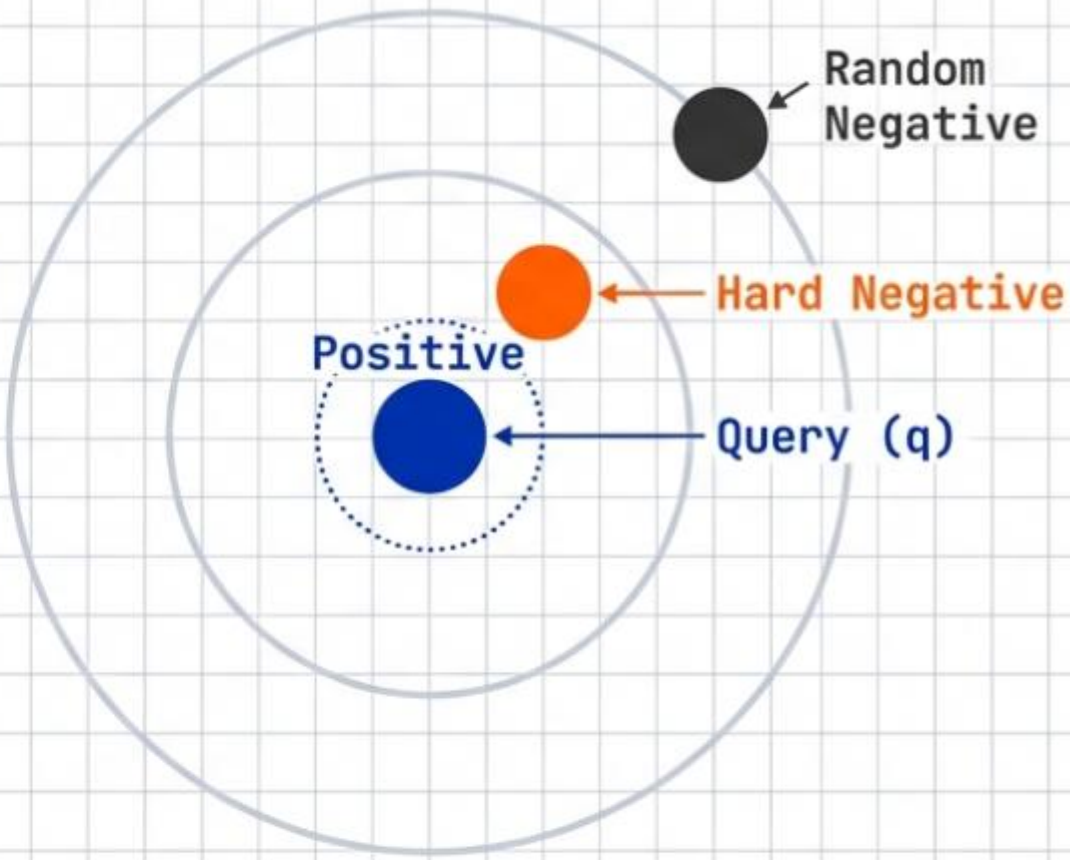



The model learns a geometric space where relevant pairs are close and irrelevant pairs are far.

$$L = -\log\left(\frac{\exp(s(q, d^+))}{\sum \exp(s(q, d^-))}\right)$$

# Hard Negatives

Random negatives are too easily distinguished. Systems must mine “hard negatives”—documents that look similar but are factually wrong.




Hard negatives force the embedding model to refine its understanding of semantic nuance.

| | |
|---|---|
| Paraphrase Equivalence | 'Reset phone' ⟷ (approx equal) 'Factory reset steps' |
| Synonymy | 'Refund' ⟷ (approx equal) 'Return' |
| Intent | Distinguishing 'Troubleshoot' vs. 'Buy' |
| Topical Relatedness | Battery ⟷ Charger ⟷ Adapter |
| Entity Semantics | Categorizing iPhone/Android as 'Devices' (→ Devices) |

**"TF-IDF represents words.**
**Dense embeddings represent meaning."**

**Lexical Search (BM25)**

Precision for exact phrases, codes, and SKUs.

+

**Dense Retrieval**

Recall for synonyms, intent, and concept matching.

=

**Hybrid Search**

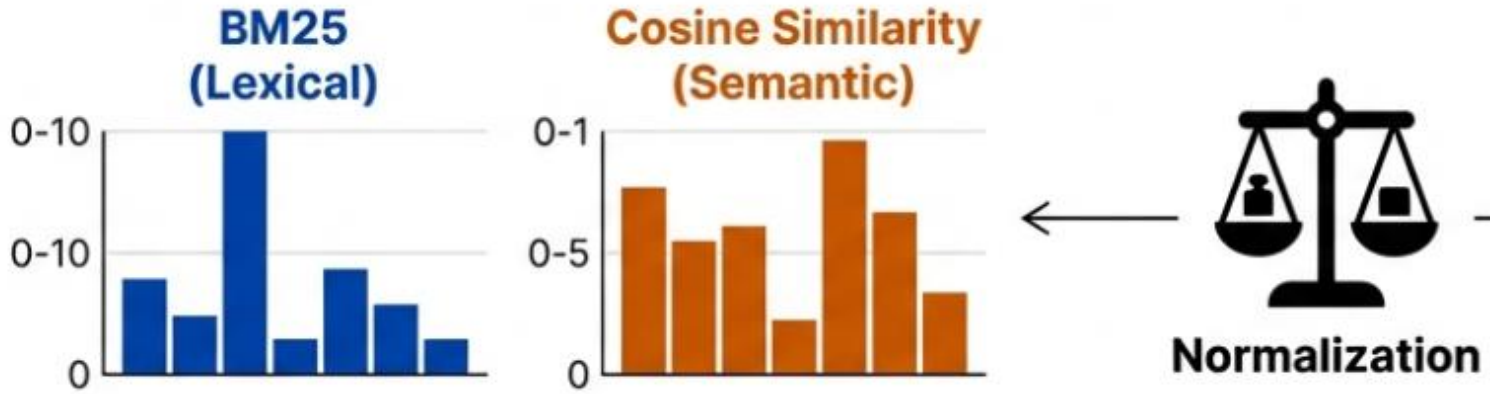
Robust retrieval across all query types.

Deploying dense retrieval alongside lexical methods ensures coverage for both semantic intent and precise keyword requirements.

# Score Fusion

$$S(d) = \alpha \cdot \text{LexicalScore} + (1 - \alpha) \cdot \text{SemanticScore}$$

Controls balance




**CRITICAL:** Must normalize scores first (e.g., Min-Max or Z-score) because BM25 and Cosine Similarity operate on different scales.

**Best For:** When you can keep score scales stable and tuned.

# Reciprocal Rank Fusion (RRF)

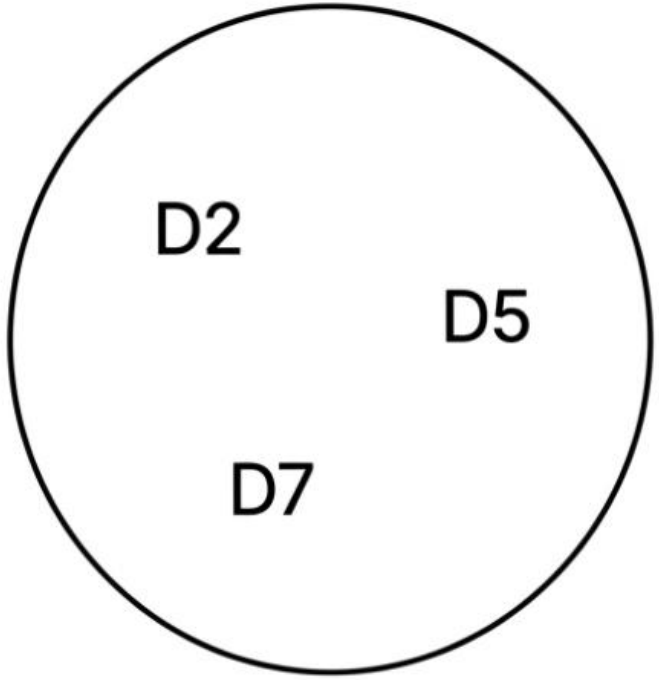
$$\mathrm{RRF}(d) = \frac{1}{k + \mathrm{Rank}_{\mathrm{Lexical}}} + \frac{1}{k + \mathrm{Rank}_{\mathrm{Semantic}}}$$

- Uses position (rank), not score value.
- k is a constant (Commonly 60 for production, 10 for teaching).
- **Logic:** If a doc is high in either list, it scores well. If high in both, it scores very well.

**Best For:** When you don't want to worry about normalizing score scales.

# Evaluating Retrieval

Ground Truth (Relevant Set)




$|R| = 3$ (Total Relevant Items)

Ranked Results (System Output)

1. D5 ✓
2. D1 ✗
3. D2 ✓
4. D9 ✗
5. D7 ✓

# Precision

Ranked Results (System Output)

1. D5 ✓
2. D1 ✗
3. D2 ✓
4. D9 ✗
5. D7 ✓

Total Retrieved = 5

**Precision = Relevant Retrieved / Total Retrieved**

**P = 3 / 5 = 0.60**

Penalizes noise (False Positives). Critical when wrong answers are costly, such as in medical policy or compliance search.

Thank You

# Appendix